TEXORA · LECTURE SHEET

Measures of Central Tendency

One number to stand for a whole distribution — the arithmetic mean by three methods, the weighted mean, median and mode for grouped data, geometric and harmonic means, and how to choose the right average.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 3 · LEVEL 2 TERM 1

1. The idea of an average

A frequency table (Topic 2) shows everything but summarises nothing. A **measure of central tendency** — an average — is a single value that represents the whole set, lying somewhere near its centre. Averages make comparison possible: two classes, two mills, two years can be compared through one figure each, where whole tables cannot be compared at a glance. But no single average tells the whole story; Topic 4 will add the spread around the centre.

A **good average** should be: (i) rigidly defined, so everyone computes the same value; (ii) based on all the observations; (iii) simple to understand and calculate; (iv) capable of further algebraic treatment; (v) not unduly affected by a few extreme values; and (vi) stable from sample to sample. No average scores perfectly on all six — the choice in practice is a compromise, set out in Section 8.

2. The arithmetic mean: direct method

The **arithmetic mean** (AM) is the sum of the values divided by their number. For ungrouped data x_1, …, x_n:

$$\bar{x} = \Sigma x / n$$

For grouped data, each class is represented by its midpoint x with frequency f:

$$\bar{x} = \Sigma fx / N, \quad N = \Sigma f$$

WORKED EXAMPLE

Shared dataset (marks of 45 students, Topic 2). The fx column: 10, 90, 250, 490, 360, 220, 65, giving Σfx = 1485. Therefore

$$\bar{x} = 1485 / 45 = 33.0 \text{ marks}$$

The average student scored 33.0 out of 70. Note the key property: deviations from the mean balance out, $\Sigma(x - \bar{x}) = 0$ — the mean is the balance point of the distribution.

3. Short-cut methods: assumed mean and step deviation

When midpoints are large or awkward, pick any convenient value A (the **assumed mean**, usually a middle midpoint) and work with deviations $d = x - A$:

$$\bar{x} = A + \Sigma fd / N \quad \text{(assumed-mean method)}$$

If the classes have equal width h, divide the deviations by h as well, writing $u = (x - A)/h$, the small whole numbers …, −2, −1, 0, 1, 2, …:

$$\bar{x} = A + (\Sigma fu / N) \times h \quad \text{(step-deviation method)}$$

WORKED EXAMPLE

Shared dataset with A = 35, h = 10. Then u = −3, −2, −1, 0, 1, 2, 3 and fu = −6, −12, −10, 0, 8, 8, 3, so Σfu = −9. Hence $\bar{x} = 35 + (-9/45) \times 10 = 35 - 2 =$ **33.0 marks** — the same mean, reached with far smaller arithmetic. Both short-cut methods are exact, not approximations.

4. Weighted mean

When values carry different importance, each value is multiplied by its **weight** w before averaging:

$$\text{Weighted mean} = \Sigma wx / \Sigma w$$

WORKED EXAMPLE

A result combines three components scoring 20, 30 and 40 with weights 1, 2 and 3 (the third counts three times as much as the first):

$$\text{Weighted mean} = (20\times1 + 30\times2 + 40\times3) / 6 = 200/6 = 33.33$$

The plain mean of 20, 30, 40 would be 30; weighting pulls the result toward the heavily weighted 40. Course grade-point averages are weighted means, with credit hours as weights.

5. The median

The **median** is the middle value when observations are arranged in order: half lie below it, half above. For ungrouped data, arrange the values in ascending order and take the middle one (for an odd count), or the average of the two middle ones (for an even count). Extreme values cannot move the median —

one very high mark leaves it unchanged — which is why the median is the preferred average for incomes and other skewed data.

For grouped data the median is found by interpolation inside the **median class**, the class whose cumulative frequency first reaches N/2:

$$\text{Median} = L + ((N/2 - cf) / f) \times h$$

where L = lower boundary of the median class, cf = cumulative frequency before it, f = its frequency, h = class width.

WORKED EXAMPLE

Shared dataset: N/2 = 22.5. Cumulative frequencies run 2, 8, 18, 32, … so the median class is 30–40 (cf reaches 32, passing 22.5). With L = 30, cf = 18, f = 14, h = 10:

$$\text{Median} = 30 + (22.5 - 18)/14 \times 10 = 30 + 3.21 = 33.21 \text{ marks}$$

Half the students scored below about 33.21 and half above — close to the mean of 33.0, as expected for a fairly symmetric distribution.

6. The mode

The **mode** is the most frequent value — the peak of the distribution. For ungrouped data it is found by inspection; a set may have one mode, two (bimodal) or none. For grouped data, interpolation inside the **modal class** (the class of highest frequency) gives:

$$\text{Mode} = L + ((f_1 - f_0) / (2f_1 - f_0 - f_2)) \times h$$

where f_1 = frequency of the modal class, f_0 and f_2 = frequencies of the classes before and after it.

WORKED EXAMPLE

Shared dataset: modal class 30–40 with $f_1 = 14$, $f_0 = 10$, $f_2 = 8$, L = 30, h = 10:

$$\text{Mode} = 30 + (14 - 10)/(2\times14 - 10 - 8) \times 10 = 30 + (4/10) \times 10 = 30 + 4.0 = 34.0 \text{ marks}$$

The mode is the business average: the most commonly demanded shoe size or shirt size is a mode, and manufacturers stock to it.

Empirical relation. For a moderately skewed distribution, Karl Pearson observed the approximate relation

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$$

Check with our results: $3 \times 33.21 - 2 \times 33.0 = 99.63 - 66.0 = 33.63 \approx 34.0$. The agreement is close but not exact — the relation is empirical, a useful check and a way to estimate one average from the other two, not a law.

7. Geometric and harmonic means; positional averages

The **geometric mean** (GM) of positive values is the n-th root of their product; the **harmonic mean** (HM) is the reciprocal of the mean of the reciprocals:

$$GM = (x_1\, x_2 \ldots x_n)^{1/n}, \quad HM = n / \Sigma(1/x)$$

WORKED EXAMPLE

For the values 2, 4, 8: AM = (2 + 4 + 8)/3 = 14/3 = **4.67**; GM = $(2 \times 4 \times 8)^{1/3} = 64^{1/3}$ = **4.00**; HM = 3 / (1/2 + 1/4 + 1/8) = 3/0.875 = **3.43**. The order illustrates the general inequality for positive data:

$$AM \geq GM \geq HM$$

with equality only when all values are equal.

When each is used. The GM suits averaging ratios, percentages and growth rates — an index that rises 10% then 21% grows at a steady GM rate, not the AM of the two. The HM suits averaging rates over equal amounts of work: average speed over equal distances, or average price when equal money (not equal quantity) is spent each time. Using the AM in either case systematically overstates the answer.

Positional averages. The median generalises to **quartiles**, which split ordered data into four equal parts: Q_1 at position (N + 1)/4 (ungrouped) in the ordered list, the median Q_2 at the halfway position, and Q_3 at three times the first position. Deciles (tenths) and percentiles (hundredths) follow the same idea. Quartiles return in Topic 4 as the basis of quartile deviation.

8. Choosing the average: comparison

AVERAGE	BASED ON	BEST USED WHEN	MAIN LIMITATION
Mean	All values	Further calculation is needed; distribution fairly symmetric	Distorted by extreme values

Median	Middle position only	Skewed data (income, prices); open-ended classes present	Ignores actual values; weak for algebra
Mode	Highest frequency	The typical or most demanded item is wanted (sizes, styles)	May not exist, or may not be unique
GM	All values (product)	Averaging ratios and growth rates	Undefined if any value is zero or negative
HM	All values (reciprocals)	Averaging speeds and rates over equal work	Strongly pulled down by small values; less intuitive

For the shared dataset the three main averages — mean 33.0, median 33.21, mode 34.0 — huddle together, the signature of a distribution close to symmetric. When they drift far apart, the drift itself is information: it signals skewness, and the median is then usually the safer representative.

Glossary

TERM	MEANING
Central tendency	The tendency of values to cluster around a central, representative value.
Arithmetic mean	The sum of the values divided by their number; $\Sigma fx/N$ for grouped data.
Assumed mean	A convenient value A used as a temporary origin to simplify mean calculations.
Weighted mean	An average in which each value is multiplied by a weight reflecting its importance: $\Sigma wx/\Sigma w$.
Median	The middle value of ordered data; half the observations lie on each side.
Median class	The class whose cumulative frequency first reaches N/2.
Mode	The most frequent value; the peak of the distribution.
Geometric mean	The n-th root of the product of n positive values; used for ratios and growth.
Harmonic mean	The reciprocal of the mean of the reciprocals; used for rates and speeds.
Quartiles	Values dividing ordered data into four equal parts: Q_1, Q_2 (median), Q_3.

Key points & summary

- A good average is rigidly defined, based on all observations where possible, simple, algebraically workable and not dominated by extremes.
- Grouped mean = Σfx/N = 1485/45 = 33.0 marks for the shared dataset; assumed-mean and step-deviation methods give the identical result with smaller numbers.
- Weighted mean = Σwx/Σw: the components 20, 30, 40 with weights 1, 2, 3 give 33.33.
- Grouped median by interpolation: 33.21 marks (median class 30–40); grouped mode: 34.0 marks (modal class 30–40).
- Empirical relation Mode ≈ 3 Median − 2 Mean gives 33.63 ≈ 34.0 here — approximate, and a good arithmetic check.
- For 2, 4, 8: AM 4.67 ≥ GM 4.00 ≥ HM 3.43. GM is for growth and ratios; HM is for speeds and rates.
- Mean, median and mode close together means near-symmetry; far apart means skewness, favouring the median.

Practice

Answer in your own words; brief model answers follow.

1. Compute the mean of the shared dataset by the step-deviation method with A = 25 and h = 10, showing the u and fu columns.
2. A student scores 50, 60 and 70 in three subjects carrying credits 2, 3 and 5. Find the weighted mean score.
3. Using the empirical relation, estimate the mode of a distribution whose mean is 40 and median is 42. Comment on the skewness.
4. A car travels a fixed route at 40 km/h and returns over the same distance at 60 km/h. Find its average speed for the round trip, and name the mean you used.
5. Production grows by 25% in one year and by 44% in the next. Find the average yearly growth rate.
6. Explain why the median, not the mean, is quoted for household income in official reports.

Answer key

1. u = (x − 25)/10 = −2, −1, 0, 1, 2, 3, 4; fu = −4, −6, 0, 14, 16, 12, 4; Σfu = 36. Mean = 25 + (36/45) × 10 = 25 + 8 = 33.0 marks, agreeing with the direct method.
2. Weighted mean = (50×2 + 60×3 + 70×5)/(2 + 3 + 5) = (100 + 180 + 350)/10 = 630/10 = 63.
3. Mode ≈ 3 × 42 − 2 × 40 = 126 − 80 = 46. Median (42) exceeds the mean (40) and the estimated mode is higher still, so the distribution is negatively skewed (a tail of low values pulls the mean down).

4. Equal distances, so use the harmonic mean: HM = 2/(1/40 + 1/60) = 2/(5/120) = 48 km/h — not the arithmetic mean 50 km/h.
5. Growth factors are 1.25 and 1.44. GM = √(1.25 × 1.44) = √1.80 = 1.342, so the average growth is about 34.2% per year. (The AM, 34.5%, slightly overstates it.)
6. Incomes are strongly skewed: a few very high incomes drag the mean far above what a typical household earns, while the median — the middle household — is unaffected by extremes and represents the typical case honestly.
